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Topic 17: Oscillations - Mark scheme

Cambridge International AS & A Level Physics (9702) | Syllabus 2025–2027

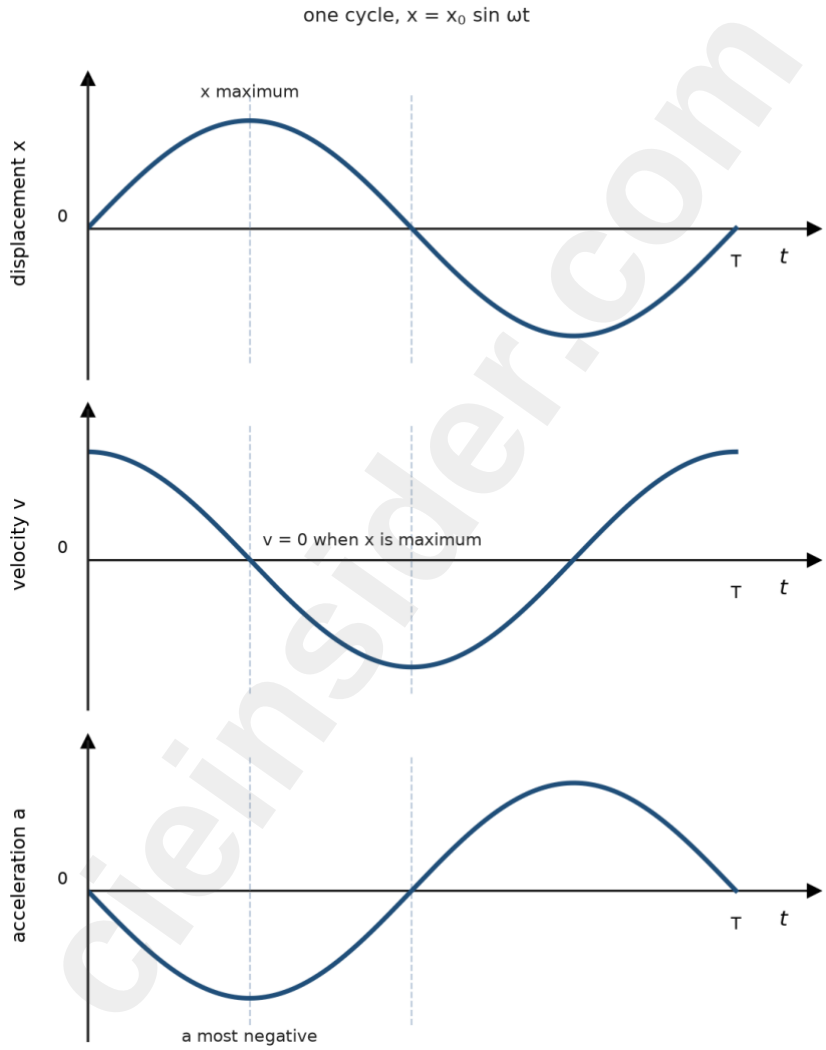
Covers: 17.1 Simple harmonic oscillations to 17.3 Damped and forced oscillations, resonance

Level: A Level

For teachers. Contains the answers to Sections 6 and 7 and model versions of the Section 4 diagrams.

Model diagrams (Section 4)

4.1 Displacement, velocity and acceleration for one cycle

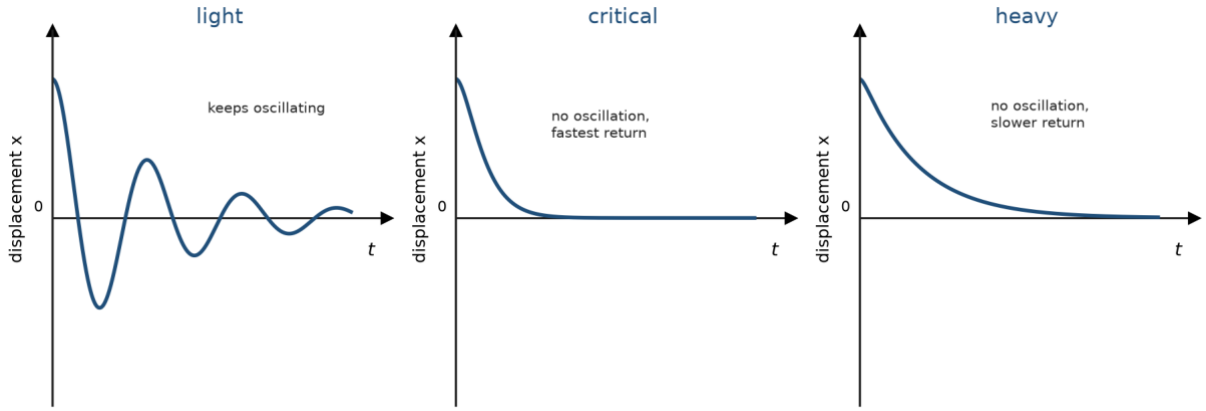


Read down the page. $x = x_0 \sin \omega t$, $v = \omega x_0 \cos \omega t$ and $a = -\omega^2 x_0 \sin \omega t$, each drawn over one cycle and scaled only so the unnumbered axes can show the shape. Where x is steepest, v is largest. Where v slopes down, a is negative.

Credit for: x starts at 0, reaches $+x_0$ at T/4, returns to 0 at T/2, reaches $-x_0$ at 3T/4, and is 0 at T; v starts at its positive maximum, is 0 at T/4 and at 3T/4, and is negative at T/2; a is 0 at t = 0, most negative when x is most positive, and 0 whenever x is 0; a is not zero at maximum displacement.

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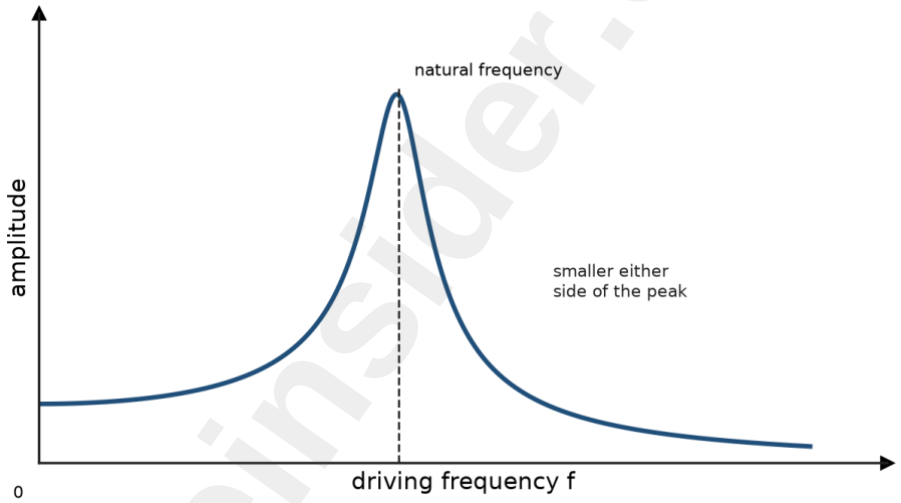
4.2 Light, critical and heavy damping



All three curves are the solution of the damped equation for release from rest at the same x_0 and the same natural frequency. Light damping (γ below ω_0) crosses equilibrium. Critical damping ($\gamma = \omega_0$) and heavy damping (γ above ω_0) do not, and the critical curve is back near zero first.

Credit for: Light: the line crosses the time axis repeatedly, with a falling amplitude; Critical: the line returns to the time axis without crossing it, and does so sooner than the heavy-damping line; Heavy: the line returns to the time axis without crossing it, more slowly than critical damping; All three start at the same positive displacement with zero gradient.

4.3 Resonance curve

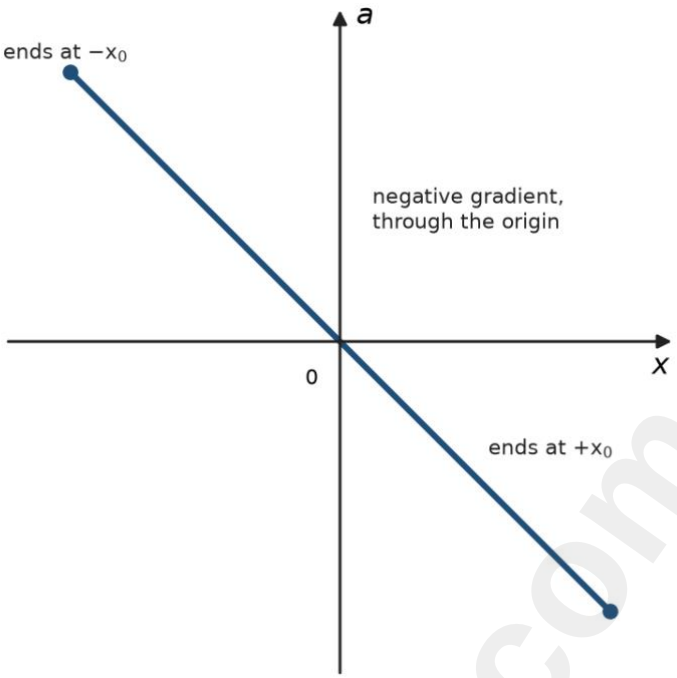


The curve is the steady-state amplitude of a driven oscillator with light damping. The peak is at the driving frequency closest to the natural frequency. Heavier damping would lower the peak; that comparison is not required by the sketch.

Credit for: Amplitude is not the same at every driving frequency; The curve has one maximum; The maximum is at the natural frequency; The amplitude is smaller both below and above that frequency.

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4.4 Acceleration against displacement



$a = -\omega^2 x$ from $x = -x_0$ to $x = +x_0$ only. The same straight line, with $\omega^2 = 50\ s^{-2}$, is the graph in Question 8, where the ends are at $x = \pm 0.080$ m and $a = \mp 4.0\ m\ s^{-2}$.

Credit for: A single straight line; The line passes through the origin; Negative gradient: a and x have opposite signs; The line stops at $+x_0$ and at $-x_0$; Not a closed loop.

Multiple-choice answers (Section 6)

Each wrong option is a specific mistake. Where an examiner report records candidates making it, the series and paper are given in brackets. Use the third column to diagnose, not just to mark.

#	Answer	Why each wrong option is wrong
1	**C**	**A** Uses ω rather than ω^2: ½ × 0.80 × 5.0 × 1.50^2 = 4.50 J **B** Does not square the amplitude: ½ × 0.80 × 5.0^2 × 1.50 = 15.0 J (Nov 2022, Paper 42) **D** Omits the factor ½: 0.80 × 5.0^2 × 1.50^2 = 45.0 J
2	**A**	**B** Uses twice the amplitude: 4.0 × 0.040 = 0.16 m s^{-1} **C** Uses $\omega^2 x_0$, which is the maximum acceleration, as if it were a speed: 4.0^2 × 0.020 = 0.32 m s^{-1} **D** Leaves the amplitude in centimetres: 4.0 × 2.0 = 8.0 m s^{-1}
3	**D**	**A** Gives the period 1/f as the angular frequency: 1/4.00 = 0.250 rad s^{-1} **B** Uses the frequency in place of the angular frequency, so the value taken is = 4.00 rad s^{-1}. (June 2025, Paper 44) **C** Uses πf instead of 2πf: π × 4.00 = 12.6 rad s^{-1}
4	**B**	**A** Uses $\omega(x_0 - x)$ and does not use the difference of squares: 4.0 × (0.50 − 0.30) = 0.80 m s^{-1} **C** Uses the maximum speed ωx_0 at a point that is not the equilibrium position: 4.0 × 0.50 = 2.00 m s^{-1} **D** Adds inside the square root: $4.0 \times \sqrt{(0.50^2 + 0.30^2)}$ = 2.33 m s^{-1}
5	**A**	**B** Uses the peak-to-peak distance 5.0 cm as the amplitude: 10^2 × 0.050 = 5.00 m s^{-2} **C** Converts centimetres to metres by dividing by 10: 10^2 × 0.25 = 25.0 m s^{-2} **D** Leaves the amplitude in centimetres: 10^2 × 2.5 = 250 m s^{-2} (June 2023, Paper 42)

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#	Answer	Why each wrong option is wrong
6	**D**	**A** Ignores the minus sign in $a = -\omega^2 x$. That minus sign is a phase difference of π rad. **B** Uses 45°, which is not the phase difference between any of x, v and a for this equation. **C** This is the phase difference between displacement and velocity, not between displacement and acceleration.
7	**C**	**A** Takes the amplitude to be the same however the system is driven. Resonance is a maximum at the natural frequency only. (June 2021, Paper 41) **B** Mixes resonance up with damping. Damping is caused by a resistive force. Resonance is a maximum amplitude of a forced oscillation. **D** Doubles the natural frequency. The maximum is at the natural frequency itself, not at $2f_0$.
8	**B**	**A** Uses T = 1/ω, which treats the angular frequency as a frequency in hertz: 1/20 = 0.0500 s (June 2025, Paper 44) **C** Inverts the factor 2π: 20/(2π) = 3.18 s **D** Multiplies by 2π instead of dividing: 2π × 20 = 126 s

Mark scheme (Section 7)

Question 1 [6]

(a) Distance (of the oscillating object) from the equilibrium position, in a stated direction **[1]**

Not displacement from an arbitrary zero. Not amplitude.

(b) Maximum displacement from the equilibrium position **[1]**

(c) Number of oscillations (or cycles) per unit time **[1]**

Not 1/T on its own, and not 'per second' unless the number of oscillations is also stated. Not the number of waves.

(d) π/2 rad (or 90°, or a quarter of a cycle). Velocity leads the displacement. Accept the phase difference without saying which leads **[1]**

(e) T = 1/4.00 = 0.250 s **[1]**

$\omega = 2\pi \times 4.00 = 25.1$ rad s^{-1} **[1]**

Question 2 [10]

(a) a is proportional to x (the factor 180 is constant) **[1]**

The minus sign means a is in the opposite direction to x **[1]**

a = 0 when x = 0, so the fixed point is the equilibrium position **[1]**

A definition of simple harmonic motion with no reference to this equation does not score.

(b) $\omega^2 = 180$ **[1]**

$\omega = \sqrt{180} = 13.4$ rad s^{-1} (accept 13.42). ω = 180 does not score the second mark: 180 is ω^2 **[1]**

Do not put the minus sign inside the square root.

(c) f = 13.4/(2π) = 2.14 Hz (accept 2.13 Hz from ω = 13.4) **[1]**

(d) 12 mm = 0.012 m **[1]**

$a_0 = 180 \times 0.012 = 2.16$ m s^{-2} **[1]**

Using 12 as if it were in metres gives 2160 m s^{-2} and does not score the value.

(e) $v_0 = \omega x_0 = 13.4 \times 0.012$ **[1]**

$v_0 = 0.161$ m s^{-1} **[1]**

Question 3 [8]

(a) $x_0 = 0.050$ m and ω = 8.0 rad s^{-1}, both required **[1]**

(b) ωt = 8.0 × 0.10 = 0.80 rad **[1]**

x = 0.050 sin 0.80 = 0.0359 m (accept 0.0358 m to 0.036 m) **[1]**

(c) $v_0 = 8.0 \times 0.050 = 0.400$ m s^{-1}, so v = 0.400 cos 0.80 **[1]**

v = 0.279 m s^{-1} **[1]**

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(d) $v = 8.0 \times \sqrt{(0.050^2 - 0.0359^2)}$ **[1]**

$v = 0.279$ m s^{-1} **[1]**

Positive, because cos ωt is positive / the particle is still moving in the positive direction (ωt = 0.80 rad is in the first quarter of the cycle) **[1]**

Question 4 [8]

(a) 4.0 cm (accept 0.040 m if the unit matches) **[1]**

(b) 0.80 s, from one positive maximum to the next, or from t = 0 to the second crossing of zero in the same direction **[1]**

(c) $\omega = 2\pi/0.80$ **[1]**

$\omega = 7.85$ rad s^{-1} **[1]**

(d) $x_0 = 0.040$ m used in $v_0 = \omega x_0$ **[1]**

$v_0 = 0.314$ m s^{-1} **[1]**

Leaving the amplitude as 4.0 cm gives 31.4 m s^{-1} and does not score the value.

(e) $a = -\omega^2 x_0 = -(7.85)^2 \times 0.040$ **[1]**

$a = -2.47$ m s^{-2} (accept magnitude 2.47 m s^{-2} if the direction is stated as towards equilibrium) **[1]**

Question 5 [9]

(a) Kinetic energy is maximum at equilibrium and zero at both extreme displacements **[1]**

Potential energy of the oscillation is zero at equilibrium **[1]**

Potential energy is maximum at both extremes, not only at the top **[1]**

Total energy is constant, and kinetic plus potential equals that total **[1]**

'Maximum at the top and minimum at the bottom' does not score the potential-energy marks.

(b) $E = \frac{1}{2} \times 0.150 \times 10^2 \times 0.040^2$ **[1]**

$E = 0.0120$ J **[1]**

Using 4.0 cm without converting gives 120 J and does not score the value.

(c) $E_K = 0.0120 - \frac{1}{2} \times 0.150 \times 100 \times 0.020^2 = 0.00900$ J, or $\frac{1}{2}m\omega^2(x_0^2 - x^2)$ **[1]**

$v = \omega\sqrt{(x_0^2 - x^2)}$ or $v = \sqrt{(2E_K/m)}$ **[1]**

$v = 0.346$ m s^{-1} **[1]**

Question 6 [6]

(a) A resistive force (friction, drag, or any named resistive force) **[1]**

(b) Light damping **[1]**

The mass keeps oscillating and the amplitude decreases **[1]**

'Damping' alone does not score the name. 'It takes a long time to stop' does not distinguish light from heavy damping.

(c) Critical damping returns to equilibrium in the shortest time without oscillating **[1]**

Heavy damping also does not oscillate, but the return is slower **[1]**

(d) Energy is removed by the resistive force and falls to the equilibrium value without the mass oscillating **[1]**

Question 7 [6]

(a) Maximum amplitude of oscillation **[1]**

when the driving frequency equals the natural frequency. Both ideas required for both marks; one idea scores one mark **[1]**

(b) 28 Hz **[1]**

(c) The resistive force dissipates energy **[1]**

The driver supplies energy at the same rate, so the amplitude can stay constant **[1]**

(d) The amplitude is smaller (than at 28 Hz) **[1]**

Question 8 [9]

(a) Straight line, so a is proportional to x **[1]**

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Line passes through the origin, so a = 0 at the equilibrium position **[1]**
Negative gradient, so a is opposite in direction to x **[1]**
A definition with no reference to the line does not score.
(b) Gradient = (−4.0 − 4.0)/(0.080 − (−0.080)) = −50 s^{-2}, so $\omega^2 = 50$ **[1]**
$\omega = \sqrt{50} = 7.07$ rad s^{-1} **[1]**
ω = 50, or a negative value from the gradient, does not score the second mark.
(c) $v_0 = \omega x_0 = 7.07 \times 0.080$ **[1]**
$v_0 = 0.566$ m s^{-1} **[1]**
(d) $E = ½ \times 0.400 \times 50 \times 0.080^2$ **[1]**
E = 0.0640 J (accept 0.064 J) **[1]**

Syllabus coverage

Where each checklist statement is practised in this booklet.

Ref	Statement	Practised in
17.1.1	understand and use the terms displacement, amplitude, period, frequency, angular frequency and phase difference in the context of oscillations, and express the period in terms of both frequency and angular frequency	Q1, MCQ3, MCQ6, MCQ8, WE1
17.1.2	understand that simple harmonic motion occurs when acceleration is proportional to displacement from a fixed point and in the opposite direction	Q2(a), Q8(a), Diagram 4.4, WE1
17.1.3	use $a = -\omega^2 x$ and recall and use, as a solution to this equation, $x = x_0 \sin \omega t$	Q2, Q3(a), (b), WE1, MCQ5
17.1.4	use the equations $v = v_0 \cos \omega t$ and $v = \pm \omega\sqrt{(x_0^2 - x^2)}$	Q3(c), (d), Q4(d), MCQ2, MCQ4, WE1, WE2
17.1.5	analyse and interpret graphical representations of the variations of displacement, velocity and acceleration for simple harmonic motion	Diagram 4.1, Q4, Q8, Diagram 4.4
17.2.1	describe the interchange between kinetic and potential energy during simple harmonic motion	Q5(a), WE2, WE3
17.2.2	recall and use $E = ½m\omega^2 x_0^2$ for the total energy of a system undergoing simple harmonic motion	Q5(b), (c), Q8(d), MCQ1, WE2, WE3
17.3.1	understand that a resistive force acting on an oscillating system causes damping	Q6(a), Q7(c)
17.3.2	understand and use the terms light, critical and heavy damping and sketch displacement–time graphs illustrating these types of damping	Diagram 4.2, Q6(b), (c)
17.3.3	understand that resonance involves a maximum amplitude of oscillations and that this occurs when an oscillating system is forced to oscillate at its natural frequency	Diagram 4.3, Q7, MCQ7

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